# Proving That a Periodic Orbit Exists

**Thursday, December 3, 2026**

**Definition: Positive semi-orbit**

Let $z(t; z_0)$ be a solution defined for every $t \geq 0$ and satisfying $z(0; z_0) = z_0$. Its **positive semi-orbit** is

$$\gamma^+(z_0) = \{z(t; z_0) : t \geq 0\}.$$

**Definition: Omega-limit set**

Let $z(t; z_0)$ be a solution defined for every $t \geq 0$ and satisfying $z(0; z_0) = z_0$. Its **omega-limit set** is

$$\omega(z_0) = \left\{ p \in \mathbb{R}^2 : \begin{array}{l} \text{there is a sequence } t_n \to \infty \\ \text{such that } z(t_n; z_0) \to p \end{array} \right\}.$$

**Proposition: Basic omega-limit properties**

Let $U \subseteq \mathbb{R}^2$ be open, let $F : U \to \mathbb{R}^2$ be locally Lipschitz, and let $z(t; z_0)$ solve $\dot{z} = F(z)$. If $\gamma^+(z_0)$ lies in a compact subset of $U$, then $\omega(z_0)$ is nonempty, compact, and invariant.

**Theorem: Poincaré–Bendixson theorem**

Let $U \subseteq \mathbb{R}^2$ be open and let $F \in C^1(U)$. Suppose the positive semi-orbit of $z_0$ is contained in a compact subset of $U$. If $\omega(z_0)$ contains no equilibrium of

$$\dot{z} = F(z),$$

then $\omega(z_0)$ is a periodic orbit.

**Recall: Positive invariance**

A set $D$ is **positively invariant** for $\dot{z} = F(z)$ if every solution satisfies

$$z(0) \in D \quad \Longrightarrow \quad z(t) \in D \quad \text{for every } t \geq 0 \text{ for which the solution is defined.}$$

**Definition: Trapping region**

For a flow on an open set $U \subseteq \mathbb{R}^2$, a **trapping region** is a nonempty, compact, connected, positively invariant set $D \subset U$.

**Observation: Inward-pointing boundary test**

Let $F$ be locally Lipschitz near a set $D$ whose boundary is piecewise continuously differentiable. At a smooth point of $\partial D$, let $n$ be the outward unit normal. The strict inequality

$$F \cdot n < 0$$

means that the field crosses into $D$. For the non-strict tangent-cone test, require $F \cdot n \leq 0$ at every smooth boundary point. At a corner, require this inequality for the outward normal of every incident smooth side. These conditions prevent a solution from leaving $D$ through its boundary.

**Corollary: Trapping-region criterion**

Let $F \in C^1(U)$ for an open set $U \subseteq \mathbb{R}^2$, and let $D \subset U$ be a trapping region. If $D$ contains no equilibrium of $\dot{z} = F(z)$, then $D$ contains at least one periodic orbit.

**Proof.** Choose $z_0 \in D$. Positive invariance keeps the solution in the compact set $D \subset U$. The continuation theorem therefore makes the solution global forward in time, and

$$\omega(z_0) \subseteq D.$$

The set $\omega(z_0)$ contains no equilibrium, so the Poincaré–Bendixson theorem makes it a periodic orbit in $D$. $\square$

**Visualization: Boundary directions for an annular trap**

For

$$D = \{(r, \theta) \in (0, \infty) \times S^1 : r_- \leq r \leq r_+\}, \qquad 0 < r_- < r_+,$$

the outward normals and inward conditions are

| boundary | outward normal | inward condition |
|---|---|---|
| $r = r_+$ | $e_r$ | $\dot{r} < 0$ |
| $r = r_-$ | $-e_r$ | $\dot{r} > 0$ |

Here $e_r$ denotes the radial unit vector pointing toward increasing $r$. On the inner boundary, “outward from the annulus” means “into the hole.”

**Question: Construct a trapping annulus**

Consider

$$\dot{r} = r(1 - r^2) + \mu r \cos\theta, \qquad \dot{\theta} = 1, \qquad \theta \in S^1, \qquad 0 < \mu < 1.$$

Prove that the system has a periodic orbit.

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**Solution.** Choose $0 < \delta < 1$ and define

$$r_- = (1 - \delta)\sqrt{1 - \mu}, \qquad r_+ = (1 + \delta)\sqrt{1 + \mu}.$$

On $r = r_-$, the bound $\cos\theta \geq -1$ gives

$$\begin{aligned}\dot{r} &= r_-(1 - r_-^2 + \mu\cos\theta) \\ &\geq r_-(1 - r_-^2 - \mu) \\ &= r_-(1 - \mu)\big[1 - (1 - \delta)^2\big] > 0.\end{aligned}$$

On $r = r_+$, the bound $\cos\theta \leq 1$ gives

$$\begin{aligned}\dot{r} &= r_+(1 - r_+^2 + \mu\cos\theta) \\ &\leq r_+(1 - r_+^2 + \mu) \\ &= r_+(1 + \mu)\big[1 - (1 + \delta)^2\big] < 0.\end{aligned}$$

Thus

$$D = \{(r, \theta) : r_- \leq r \leq r_+,\ \theta \in S^1\}$$

is a trapping annulus. The map

$$(r, \theta) \longmapsto (r\cos\theta, r\sin\theta)$$

identifies $(0, \infty) \times S^1$ with the punctured plane, so $D$ is a planar annulus and the planar trapping-region criterion applies. Since $\dot{\theta} = 1$, the system has no equilibrium in $D$. The trapping-region criterion gives at least one periodic orbit in $D$. It gives no uniqueness or stability conclusion.

**Question: Construct a polygonal trap for the glycolysis model**

For

$$\dot{x} = -x + ay + x^2 y, \qquad \dot{y} = b - ay - x^2 y, \qquad a, b > 0,$$

construct a compact positively invariant polygon in the closed nonnegative quadrant $\{(x, y) : x \geq 0,\ y \geq 0\}$.

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**Solution.** Choose

$$Y > \frac{b}{a}$$

and let $D$ have vertices

$$(0, 0), \qquad (b + Y, 0), \qquad (b, Y), \qquad (0, Y).$$

**Question: Construct a polygonal trap for the glycolysis model (continued)**

Its four sides satisfy:

| side | calculation | direction |
|---|---|---|
| $x = 0$ | $\dot{x} = ay \geq 0$ | right |
| $y = 0$ | $\dot{y} = b > 0$ | up |
| $y = Y$ | $\dot{y} = b - (a + x^2)Y \leq b - aY < 0$ | down |
| $x + y = b + Y$ | $\dot{x} + \dot{y} = b - x \leq 0$ | toward $x + y \leq b + Y$ |

On the diagonal, $x \geq b$. At its top endpoint $x = b$, the diagonal derivative is zero but the top-side inequality is strict. Thus the inward inequality holds for every side incident to each boundary point, so $D$ is positively invariant.

**Recall: Hyperbolic source**

An equilibrium of a planar system is a **hyperbolic source** if both eigenvalues of its Jacobian have positive real parts.

**Question: When does the glycolysis equilibrium repel?**

For

$$\dot{x} = -x + ay + x^2 y, \qquad \dot{y} = b - ay - x^2 y, \qquad a, b > 0,$$

find the equilibrium and determine when it is a hyperbolic source.

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**Solution.** Let

$$F(x, y) = \left(-x + ay + x^2 y,\ b - ay - x^2 y\right).$$

Adding the equations gives

$$\dot{x} + \dot{y} = b - x.$$

Thus every equilibrium satisfies $x_* = b$, and substitution gives the unique equilibrium

$$z_* = (x_*, y_*) = \left(b, \frac{b}{a + b^2}\right).$$

The Jacobian is

$$DF(x, y) = \begin{pmatrix} -1 + 2xy & a + x^2 \\ -2xy & -(a + x^2) \end{pmatrix}.$$

At $z_*$, the determinant and trace of the Jacobian are

$$\Delta = \det DF(z_*) = a + b^2 > 0$$

and

$$\tau = \operatorname{tr} DF(z_*) = -\frac{b^4 + (2a - 1)b^2 + a + a^2}{a + b^2}.$$

Let $q = b^2$. The condition $\tau > 0$ is

$$q^2 + (2a - 1)q + a + a^2 < 0.$$

The roots of the quadratic are

$$q_\pm = \frac{1 - 2a \pm \sqrt{1 - 8a}}{2}.$$

**Question: When does the glycolysis equilibrium repel? (continued)**

Therefore $z_*$ is a hyperbolic source when

$$0 < a < \frac{1}{8}, \qquad q_- < b^2 < q_+.$$

**Definition: Strict exit neighborhood**

Let $U \subseteq \mathbb{R}^2$ be open, let $F : U \to \mathbb{R}^2$ be continuous, and let $z_* \in U$ be an equilibrium of $\dot{z} = F(z)$. A compact neighborhood $E \subset U$ of $z_*$ with smooth boundary is a **strict exit neighborhood** if

$$F(z) \cdot n_E(z) > 0 \qquad (z \in \partial E),$$

where $n_E$ is the outward unit normal of $E$.

**Lemma: Lyapunov matrix for a repelling linear system**

Let $A$ be a real $2 \times 2$ matrix whose eigenvalues have positive real parts. Define

$$P = \int_0^\infty e^{-A^\mathsf{T} s} e^{-As}\, ds.$$

The integral converges to a symmetric positive definite matrix satisfying

$$A^\mathsf{T} P + PA = I.$$

Here $I$ is the $2 \times 2$ identity matrix.

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**Proof.** The eigenvalues of $-A$ have negative real parts, so $e^{-As}$ decays exponentially and the integral converges. For every $\xi \neq 0$,

$$\xi^\mathsf{T} P \xi = \int_0^\infty \|e^{-As}\xi\|^2\, ds > 0 \qquad (\xi \neq 0).$$

Moreover,

$$\begin{aligned} A^\mathsf{T} P + PA &= -\int_0^\infty \frac{d}{ds}\left(e^{-A^\mathsf{T} s} e^{-As}\right) ds \\ &= I. \end{aligned}$$

$\square$

**Proposition: Quadratic exit neighborhood of a planar hyperbolic source**

Let $U \subseteq \mathbb{R}^2$ be open, let $F \in C^2(U)$, and let $z_* \in U$ be an equilibrium of $\dot{z} = F(z)$. Suppose every eigenvalue of

$$A = DF(z_*)$$

has positive real part. Define

$$P = \int_0^\infty e^{-A^\mathsf{T} s} e^{-As}\, ds$$

and

$$V(z) = (z - z_*)^\mathsf{T} P (z - z_*).$$

Then there are $r_0 > 0$ and $\rho_0 > 0$ such that

$$\dot{V}(z) > 0 \qquad \text{whenever } 0 < \|z - z_*\| < r_0,$$

and $\{z : V(z) \le \rho\}$ is a strict exit neighborhood for every $0 < \rho \le \rho_0$.

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**Proof.** Set

$$\xi = z - z_*.$$

Taylor's theorem gives constants $C > 0$ and $\eta > 0$ and a remainder $R(\xi)$ such that

$$F(z_* + \xi) = A\xi + R(\xi), \qquad \|R(\xi)\| \le C\|\xi\|^2 \quad \text{when } \|\xi\| < \eta.$$

Hence, for some constant $C_1 > 0$,

$$\begin{aligned} \dot{V} &= 2\xi^\mathsf{T} P F(z_* + \xi) \\ &= \xi^\mathsf{T}(A^\mathsf{T} P + PA)\xi + 2\xi^\mathsf{T} P R(\xi) \\ &\ge \|\xi\|^2 - C_1\|\xi\|^3. \end{aligned}$$

Therefore, if

$$0 < \|\xi\| < \min\left\{\eta, \frac{1}{2C_1}\right\},$$

then

$$\dot{V} \ge \frac{1}{2}\|\xi\|^2 > 0.$$

Set

$$r_0 = \min\left\{\eta, \frac{1}{2C_1}\right\},$$

decreasing $r_0$ if necessary so that the closed ball of radius $r_0$ about $z_*$ lies in $U$. Because $P$ is positive definite, there is $\rho_0 > 0$ such that every set $V \le \rho$, with $0 < \rho \le \rho_0$, is a compact ellipsoid contained in that ball. On $V = \rho$, its outward unit normal is

$$n = \frac{\nabla V}{\|\nabla V\|},$$

so

$$F \cdot n = \frac{\dot{V}}{\|\nabla V\|} > 0.$$

Hence $V \le \rho$ is a strict exit neighborhood. $\square$

**Question: Force a periodic orbit in the glycolysis model**

Assume

$$\dot{x} = -x + ay + x^2 y, \qquad \dot{y} = b - ay - x^2 y,$$

where

$$0 < a < \frac{1}{8}, \qquad b > 0, \qquad q_- < b^2 < q_+, \qquad q_\pm = \frac{1 - 2a \pm \sqrt{1-8a}}{2}.$$

Let $Y > b/a$, and let $D$ be the positively invariant polygon with vertices

$$(0,0), \quad (b+Y, 0), \quad (b, Y), \quad (0, Y).$$

Prove that the system has a periodic orbit.

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**Solution.** Let

$$F(x,y) = \big(-x + ay + x^2 y,\ b - ay - x^2 y\big).$$

The unique equilibrium is

$$z_* = \left(b, \frac{b}{a+b^2}\right),$$

and it is a hyperbolic source. Set

$$A = DF(z_*), \qquad P = \int_0^\infty e^{-A^\mathsf{T} s} e^{-As}\, ds,$$

and

$$V(z) = (z - z_*)^\mathsf{T} P (z - z_*).$$

Because

$$0 < \frac{b}{a+b^2} < \frac{b}{a} < Y,$$

the point $z_*$ satisfies all four defining inequalities for $D$ strictly: $x_* > 0$, $y_* > 0$, $y_* < Y$, and $x_* + y_* < b + Y$. Hence $z_* \in \operatorname{int}(D)$.
Choose $\rho > 0$ so small that the hyperbolic-source proposition makes

$$E_\rho = \{z \in \mathbb{R}^2 : V(z) \le \rho\}$$

a strict exit neighborhood and $E_\rho$ lies in the interior of the polygonal trap $D$. Define

$$D_\rho = D \setminus \operatorname{int}(E_\rho).$$

On the inner boundary $\partial E_\rho$, the outward unit normal of $D_\rho$ is

$$n_{D_\rho} = -\frac{\nabla V}{\|\nabla V\|}.$$

Consequently,

$$F \cdot n_{D_\rho} = -\frac{\dot{V}}{\|\nabla V\|} < 0.$$

The polygon $D$ and ellipsoid $E_\rho$ are convex, and $E_\rho$ lies strictly inside $D$. Therefore $D_\rho$ is a compact, connected annular region. On its outer boundary, the positive-invariance inequalities for $D$ point inward or tangent to $D_\rho$; on its inner boundary, the strict inequality above points inward to $D_\rho$. Uniqueness therefore makes $D_\rho$ positively invariant. Thus $D_\rho$ is a compact trapping region. It contains no equilibrium, so the Poincaré–Bendixson theorem gives at least one periodic orbit in $D_\rho$.

**Question: Check the standard glycolysis parameters**

For
$$\dot{x} = -x + ay + x^2y, \qquad \dot{y} = b - ay - x^2y,$$
let
$$a = 0.08, \qquad b = 0.6.$$
Classify the equilibrium and state the conclusion justified by an annular trapping region.

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**Solution.** Let
$$F(x, y) = \Big(-x + ay + x^2y,\ b - ay - x^2y\Big).$$
The unique equilibrium is
$$z_* = \left(b, \frac{b}{a + b^2}\right) = \left(0.6, \frac{15}{11}\right).$$
The determinant of its Jacobian is
$$\Delta = \det DF(z_*) = a + b^2 = 0.08 + 0.36 = 0.44.$$
Also,
$$\begin{aligned} b^4 + (2a - 1)b^2 + a + a^2 &= 0.1296 - 0.3024 + 0.08 + 0.0064 \\ &= -0.0864, \end{aligned}$$
so
$$\tau = \operatorname{tr} DF(z_*) = \frac{0.0864}{0.44} \approx 0.1964 > 0.$$
Finally,
$$\tau^2 - 4\Delta < 0.$$
The equilibrium is an unstable spiral. The annular trapping region proves that at least one periodic orbit exists. It does not prove that the orbit is unique, isolated, or stable.

**Observation: Planar restriction**

The Poincaré–Bendixson conclusion is special to planar flows; it does not extend to autonomous systems in dimensions three and higher.

**Visualization: What the standard planar criteria establish**

| criterion | conclusion |
|---|---|
| $\dot{V} < 0$ on every non-equilibrium state | no periodic orbit |
| $\nabla\cdot(BF)$ has one strict sign on a simply connected domain | no periodic orbit in that domain |
| compact trap with no equilibrium | at least one periodic orbit |
| Poincaré–Bendixson alone | no conclusion about uniqueness or stability |